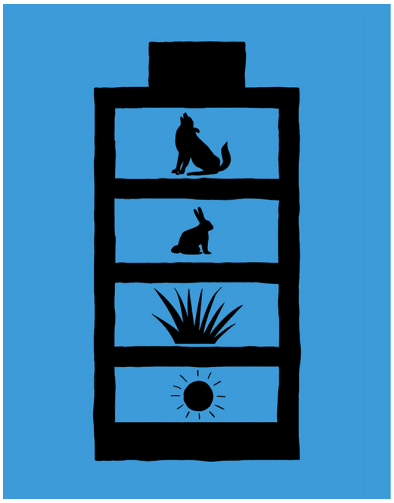


LIFE IS SUPPORTED BY A VAST NETWORK OF PROCESSES

ENERGY FLOW THROUGH ECOSYSTEMS

IN CONTEXT

KEY FIGURE

Raymond Lindeman (1915–42)

BEFORE

1913 American zoologist Victor Shelford produces the first illustrated food webs.

1920 Frederic Clements describes how groups of plant species are associated in communities.

1926 Russian geochemist Vladimir Vernadsky sees that chemicals are recycled between living and nonliving things.

1935 Arthur Tansley develops the concept of the ecosystem.

AFTER

1957 American ecologist Eugene Odum uses radioactive elements to map food chains.

1962 Rachel Carson draws attention to the accumulation of pesticides in food chains, in her book *Silent Spring*.

In 1941, Raymond Lindeman submitted the final chapter of his Ph.D. thesis for publication in the prestigious journal *Ecology*. Titled “The Trophic-